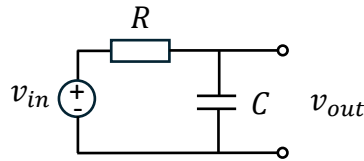


1. For the following circuit with $R = 1\ \Omega$ and $C = 1\ F$. Please plot the system response v_{out} with the following four case inputs:
 - 1). $v_{in} = 3.3 \sin(2\pi t)$;
 - 2). $v_{in} = 2 \sin(10 * 2\pi t)$;
 - 3). $v_{in} = 4 \sin(100 * 2\pi t)$;
 - 4). $v_{in} = 3.3 \sin(2\pi t) + 4 \sin(100 * 2\pi t)$; Please explain the phenomenon you observed.
 - 5). Please plot the system gain as a function of the input frequency.



In the complex frequency domain (s -domain), the transfer function $H(s)$ of the system is:

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{\frac{1}{sC}}{R + \frac{1}{sC}} = \frac{1}{1 + sRC}$$

Given that $R = 1\ \Omega$ and $C = 1\ F$, the time constant is $\tau = RC = 1\ s$.

The transfer function simplifies to:

$$H(s) = \frac{1}{1 + s}$$

Converting this to the frequency response function (let $s = j\omega$):

$$H(j\omega) = \frac{1}{1 + j\omega}$$

Therefore, the magnitude and phase characteristics of the system are:

- **Magnitude (Gain):** $|H(j\omega)| = \frac{1}{\sqrt{1+\omega^2}}$
- **Phase:** $\angle H(j\omega) = -\arctan(\omega)$

Since the input signal is in the form $A \sin(\omega t)$, the steady-state output signal is $A|H(j\omega)| \sin(\omega t + \angle H(j\omega))$. Note that the angular frequency in the problem is $\omega = 2\pi f$.

1) to 3) Theoretical Responses for Each Input Case

1) $v_{in} = 3.3 \sin(2\pi t)$

- **Frequency** $f = 1$ Hz, **Angular Frequency** $\omega = 2\pi \approx 6.28$ rad/s
- **Gain:** $|H(j2\pi)| = \frac{1}{\sqrt{1+(2\pi)^2}} \approx 0.157$
- **Phase:** $\angle H(j2\pi) = -\arctan(2\pi) \approx -1.413$ rad
- **Steady-state Output:** $v_{out1}(t) \approx 0.518 \sin(2\pi t - 1.413)$

2) $v_{in} = 2 \sin(10 \cdot 2\pi t)$

- **Frequency** $f = 10$ Hz, **Angular Frequency** $\omega = 20\pi \approx 62.83$ rad/s
- **Gain:** $|H(j20\pi)| = \frac{1}{\sqrt{1+(20\pi)^2}} \approx 0.0159$
- **Phase:** $\angle H(j20\pi) = -\arctan(20\pi) \approx -1.555$ rad
- **Steady-state Output:** $v_{out2}(t) \approx 0.0318 \sin(20\pi t - 1.555)$

3) $v_{in} = 4 \sin(100 \cdot 2\pi t)$

- **Frequency** $f = 100$ Hz, **Angular Frequency** $\omega = 200\pi \approx 628.3$ rad/s
- **Gain:** $|H(j200\pi)| = \frac{1}{\sqrt{1+(200\pi)^2}} \approx 0.00159$
- **Phase:** $\angle H(j200\pi) = -\arctan(200\pi) \approx -1.569$ rad
- **Steady-state Output:** $v_{out3}(t) \approx 0.00636 \sin(200\pi t - 1.569)$

4) Composite Input and Phenomenon Explanation

Input: $v_{in} = 3.3 \sin(2\pi t) + 4 \sin(100 \cdot 2\pi t)$

According to the **superposition principle** of linear systems, the output is the superposition (sum) of Case 1 and Case 3:

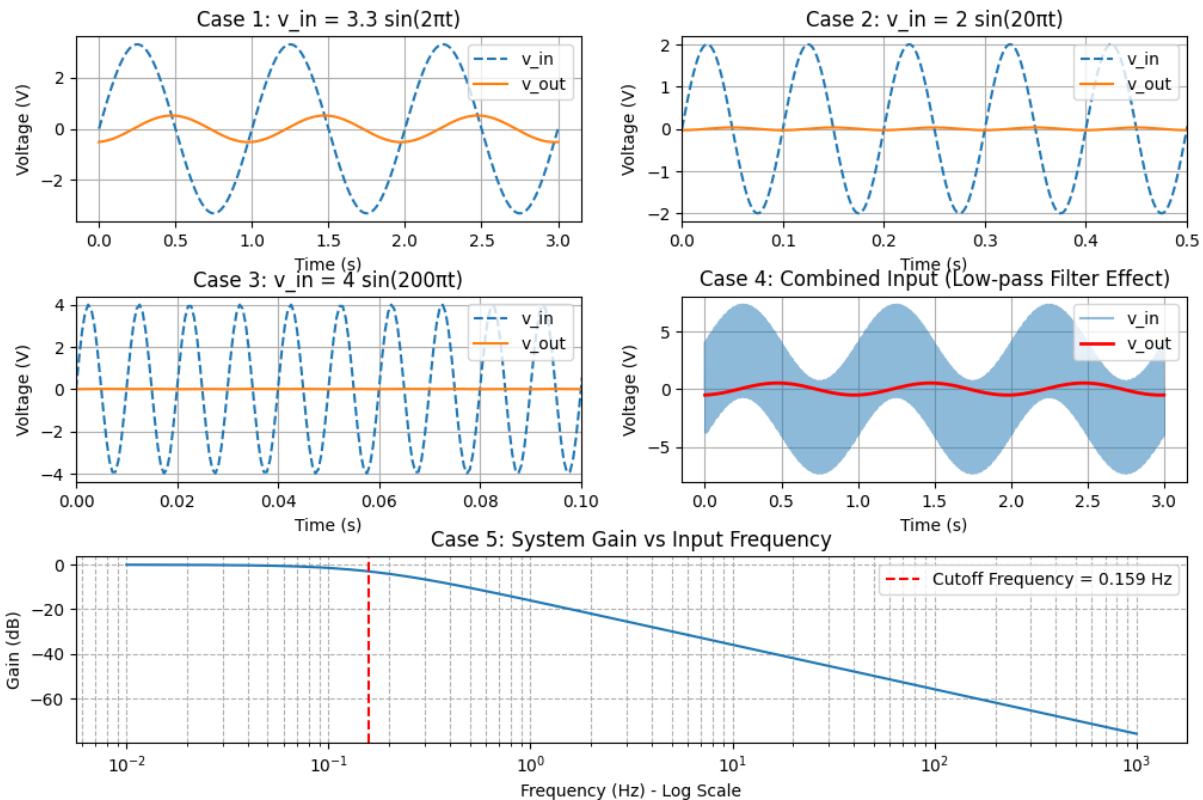
$$v_{out4}(t) \approx 0.518 \sin(2\pi t - 1.413) + 0.00636 \sin(200\pi t - 1.569)$$

Explanation of Observed Phenomena

This is a classic **Low-pass filtering** phenomenon.

- The **cutoff frequency** of this RC circuit is $\omega_c = \frac{1}{RC} = 1$ rad/s (which is $f_c \approx 0.159$ Hz)
- Although the **1 Hz low-frequency signal** undergoes some attenuation (amplitude drops from **3.3** to **0.518**), it still retains most of its energy.
- The **100 Hz high-frequency signal**, on the other hand, is greatly weakened (amplitude plummets from **4** to **0.00636**, approaching zero).

- At the output terminal, the high-frequency "glitches" or "noise" almost completely disappear, and the system outputs only the smooth, low-frequency primary waveform.



- A signal of $\sin 10\pi t$ is interference by a noise of $0.5 \sin 300\pi t$. The details can be obtained by the following python code.

```
sample_rate = 400
N = 200 # Number of sample points
T = 1 / sample_rate # sample spacing
t = np.linspace(0.0, N*T, N)
f1 = 5
f2 = 150
signal_input = np.sin(f1 * 2.0*np.pi*t)
noise = 0.5*np.sin(f2 * 2.0*np.pi*t)
measurement = signal_input+noise
```

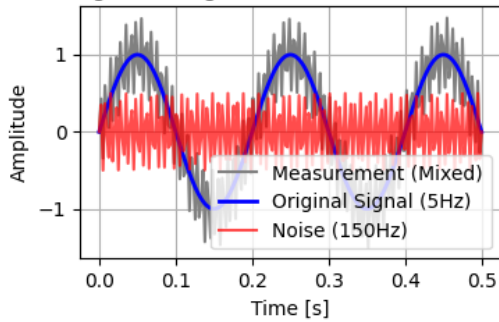
Please do FFT analysis and get the frequency response (more details can be found [here](#).) Please also use a Butterworth filter with cut off frequency of 50 Hz to filter the high frequency noise. Please plot the signal, noise, and measurement in Sub-figure 1. The FFT result in Sub-figure 2. The FFT result in dB scale in Sub-figure 3. The measurement, the signal, and the result after low pass filter in Sub-figure 4. Please also do a FFT with its original equation:

The FFT $y[k]$ of length N of the length- N sequence $x[n]$ is defined as

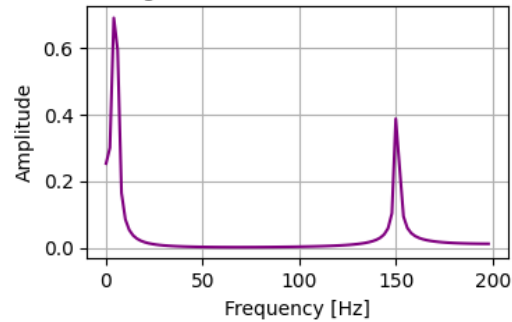
$$y[k] = \sum_{n=0}^{N-1} e^{-2\pi j \frac{kn}{N}} x[n],$$

And plot the result in Sub-figure 5. Please plot the Gain of the low-pass filter as well as the FFT of the measurement in Sub-figure 6. I have done it, and the figures are shown below for your reference. You can find the example code in the slides and if there is anything that you don't understand, please search it online.

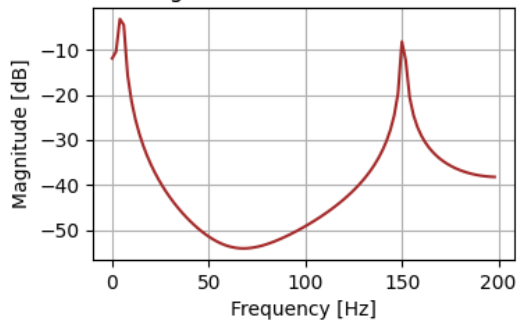
Sub-figure 1: Signal, Noise, and Measurement



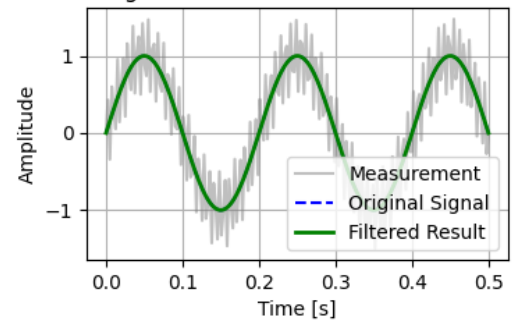
Sub-figure 2: FFT Result (Linear Scale)



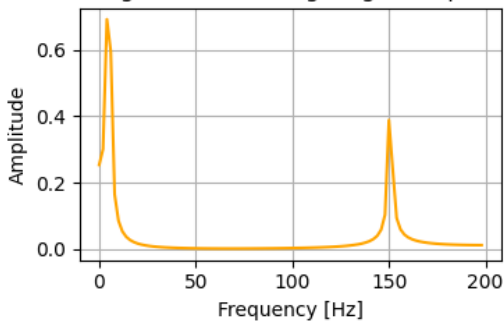
Sub-figure 3: FFT Result (dB Scale)



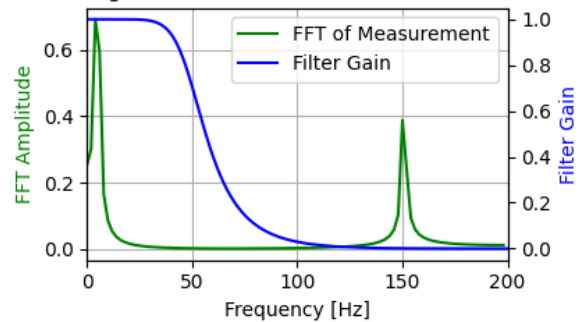
Sub-figure 4: Results After Low-Pass Filter



Sub-figure 5: FFT using Original Equation



Sub-figure 6: Filter Gain & Measurement FFT



Problem 3:

Table Prob. 1 gives 50 sample values of a random signal.

- Estimate the mean and standard deviation of the signal.
- Using an interval $\Delta y = 0.5$ V, draw the P_j and C_j discrete probability distributions for the signal.

-0.59	1.02	-0.25	-0.34	0.95	1.24	-0.30	0.21	-0.89
-1.00	1.36	0.03	0.04	-0.13	-0.71	-1.23	0.03	-1.00
0.65	0.11	0.99	0.17	0.39	2.61	-0.08	-0.33	0.99
2.15	0.91	0.89	1.43	-1.69	-0.25	2.47	-1.97	-2.26
0.42	0.05	0.26	0.33	-0.42	0.79	-0.07	-0.32	-0.66
-0.63	-0.06	-0.61	0.77	1.90				

- Mean and Standard Deviation

$$\sum y_i = 7.37$$

Mean, $\bar{y}$

$$\bar{y} = \frac{1}{N} \sum_{i=1}^{50} y_i = \frac{7.37}{50} = \mathbf{0.1474 \text{ V}}$$

Standard Deviation, s

$$s = \sqrt{\frac{1}{N-1} \sum_{i=1}^{50} (y_i - \bar{y})^2}$$

$$s \approx \mathbf{1.03 \text{ V}}$$

-

